

Cédric Nathanaël Rossboth

Computational Chemist
Molecular Simulation & ML



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What I bring

Enhanced sampling

Microsecond protein–ligand trajectories, funnel metadynamics and thermodynamic integration on a national GPU supercomputer.

Simulation engineering

Built PLUMED 2.9 with its funnel module, patched GROMACS 2023, and automated restarts and analysis with SLURM job arrays.

ML for simulation

Equivariant ML potentials replacing DFT force calls in transition-state searches across 208 alloy surfaces.

Education

Master's in Molecular and Biological Chemistry, EPFL

09/25 – 2027

- GPA 5.65 / 6.00 after year 1 · Catalysis & Sustainability · Minor in Engineering for Sustainability

Bachelor's in Chemistry and Chemical Engineering, EPFL

2022 – 2025

- GPA 5.47 / 6.00

Swiss Maturité, Lycée Blaise-Cendrars, La Chaux-de-Fonds

2022

- Biology and Chemistry, with honors

Technical skills

Simulation engines	GROMACS 2023, PLUMED 2.9 (built from source, patched), AmberTools · ligand parameterization (GAFF2/RESP)
Free-energy methods	Well-tempered & funnel metadynamics with reweighting · thermodynamic integration · MM/PB(GB)SA · nudged elastic band · convergence analysis
Workflow & HPC	SLURM job arrays and automated restarts on CSCS Alps (GH200) · Python (advanced), Bash / Linux, Git / GitHub · RDKit, pandas, NumPy, Matplotlib
Machine learning	PyTorch, scikit-learn · equivariant ML potentials (EquiformerV2/OC20) · GNNs (MPNN, GAT) · contrastive learning · symbolic regression
Electronic structure	Gaussian 16, Psi4 · DFT & post-HF · transition-state search, IRC

Research projects

Enhanced-sampling free energies for FABP4–PFAS binding

06/26 – 09/26

Laboratory of Computational Chemistry and Biochemistry, EPFL (12 ECTS) · [GitHub](#)

- Compared end-point (MM/PBSA), constrained thermodynamic integration and well-tempered funnel metadynamics with reweighting.
- Compiled PLUMED 2.9 with the funnel module and patched GROMACS 2023; automated restarts and analysis with SLURM on CSCS Alps.
- Perfluorination gives no measurable affinity gain for C10 chains; traced a TI anomaly to a pocket conformational change.

ML-potential screening of Cu-based bimetallic electrocatalysts

08/25 – 09/25

Laboratory of Artificial Chemical Intelligence, EPFL (6 ECTS) · [GitHub](#)

- Coupled an equivariant ML potential (EquiformerV2, OC20) with climbing-image transition-state search across 208 alloy surfaces, at a fraction of DFT cost.
- A lead towards more sustainable butadiene and ethylene electrosynthesis; flagged where ML barriers deviate from DFT literature.

Biocatalytic synthesis of non-natural amino acids *upcoming*

12/26 – 01/27

Laboratory of Inorganic Synthesis and Catalysis, EPFL · interdisciplinary sustainability project (10 ECTS)

- Wet-lab project designing enzymatic routes to non-natural amino acids, extending my computational background into experimental catalysis.

ML prediction of electron correlation energies

02/26 – 06/26

AI for Chemistry course, EPFL · team of 4 (3 ECTS)

- MLPs, message-passing and edge-aware GAT models predicting CC / MP2 correlation energies from HF features; chemical accuracy on benchmarks.
- My part: literature review and the interactive app for the trained models.

Symbolic regression of spin-splitting energies in Fe complexes

04/24 – 06/24

Laboratory for Computational Molecular Design, EPFL · pair project (3 ECTS) · [GitHub](#)

- Evolved interpretable genetic-programming models linking metal–ligand geometry to spin splitting; built the 150-complex DFT test set (B3LYP*).

Work experience

Machine Learning Engineer — STRUCTURE project

09/26 – now

EPFL AI Team (10 h/week)

- Contrastive models (Set Transformer + frozen language model, InfoNCE) retrieving molecular structures from $^1\text{H}/^{13}\text{C}$ NMR, HSQC, IR and MS/MS spectra.

Teaching Assistant

02/26 – now

EPFL & University of Lausanne (~9 h/week)

- BIO-707 doctoral course (45 h): turning sessions on figures, storytelling and posters into a reusable course guide.
- First-year general chemistry lab supervision and grading (UNIL).

Webmaster

09/25 – now

AdEC, Chemistry Students' Association of EPFL (~4 h/week)

- Rebuilt the association website from scratch, with a shared student resource drive and a sponsor page.

Languages

French native

English C1

German B2

Italian A2

Interests

Ice dancing: competitive 2017–2023 · 2nd at the Swiss Championships (2021, 2022) · Prix Romand 2023, for the athletes who best represented French-speaking Switzerland

Teaching: private tutoring and science communication · **Also:** outdoor calisthenics, guitar, language learning

Personal details

Age

22 years old

Nationality

Swiss and Austrian

Marital status

Single

Driving license

Swiss category B

References available on request